UKIEPC 2026

Hacking Session 1

KCPC

21 October 2026

Problems and Materials

The UKIEPC 2026 problem sheet for KCPC Year 2 Hacking Session 1.

Scan the QR code to open the official problem booklet.

`https://github.com/orthant-kcpc/kcpc-workshops/blob/main/KCPC%20Year%202%20Workshops/Hacking%20Session%201/Hacking%20Session%201%20Problems.pdf`



Problem E: Elderberry Wine

Problem: Find the total ingredients to order so that the stock can be brewed in the required $12 : 12 : 40$ ratio.

Solution:

- → Every initial ingredient is eventually used; the limiting ingredient determines $M = \max(a/12, b/12, c/40)$.
- → The final stock must be $(12M, 12M, 40M)$, so order each positive deficit.
- → Use extended precision, or defer division, for inputs up to 10^{16}.

Time complexity: $O(1)$

Problem A: Austere Sign

Problem: Rewrite the sign using at most k distinct letters, changing as few letters as possible without changing any word lengths.

Solution:

- → Count each letter's frequency across all words in linear time.
- → Keep the k most frequent letters to minimise replacements.
- → Replace every other letter with a kept letter.

Time complexity: $O(N)$

Problem I: Inverse Sum

Problem: Find distinct positive integers a, b with minimum sum such that $1/n = 1/a + 1/b$.

Solution:

- → Rearranging gives $(a - n)(b - n) = n^2$.
- → Set $x = a - n$ and $y = b - n$; minimise $x + y$ over distinct factor pairs of n^2.
- → Factor n in $O(\sqrt{n})$ and choose the largest divisor $x < n$ of n^2, with $y = n^2/x$.

Time complexity: $O(\sqrt{n} + d(n^2))$

Problem C: Cryptlet

Problem: Assign distinct digits to the five letters so that the given equal-length addition $X + Y = Z$ is valid.

Solution:

- → Only $10!/5! = 30{,}240$ distinct digit assignments need testing.
- → Precompute each letter's positional-weight sum modulo large primes in $O(|X|)$.
- → Check each assignment with constant-time modular combinations and verify candidates with two additional primes.

Time complexity: $O(|X| + 30{,}240)$

Problem K: Kilt Quilt

Problem: Find a shortest sequence of full-row and full-column brush strokes that paints a given canvas.

Solution:

- → Every cell must be covered; assume without loss of generality that all rows are painted.
- → Any untouched column reveals the required colour of every row stroke.
- → A mismatch forces row/column stroke precedence; topologically sort these constraints and report impossible if they cycle.

Time complexity: $O(N^3)$

Problem F: Footnotes

Problem: Move exactly two-thirds of the letters in a nested-bracket string into two equally sized footnotes.

Solution:

- → Let L be the total number of letters; if $3 \nmid L$, the answer is impossible.
- → Parse the brackets into a syntax tree and count letters in every subtree in $O(|s|)$.
- → Check for two disjoint $L/3$ subtrees, or an $L/3$ subtree nested inside a $2L/3$ subtree.

Time complexity: $O(|s|)$

Problem B: Bust Blasting

Problem: Find the volume of the 3D shape that remains after intersecting the statue's front, side, and top projections.

Solution:

- → Coordinate-compress each axis, leaving at most 200 intervals per axis.
- → Rasterise each 2D projection as a 200×200 mask using the even/odd fill rule.
- → Sum $\Delta x\, \Delta y\, \Delta z$ for each cuboid active in all three masks.

Time complexity: $O(N^3)$, optimisable to $O(N^2)$

Problem G: Gloucester Somersault

Problem: Count the full revolutions needed for a rolling polygonal cheese to cross the finish line while touching the ground.

Solution:

- → Interior vertices never touch the ground, so the rolling distance is determined by the convex hull.
- → One full revolution advances the cheese by the convex-hull perimeter P.
- → From the initial rightmost ground contact x_0, round up $(m - x_0)/P$; the official notes warn strict-boundary cases were absent from contest tests.

Time complexity: $O(N \log N)$

Problem L: Lawn Stripes

Problem: Route a roller over a square lawn so every cell ends with its requested cardinal direction.

Solution:

- → Give each cell one outgoing edge in its required direction; the lawn becomes a functional graph.
- → Following edges from the start reaches a cycle or boundary sink; reverse edges to form trees rooted there.
- → DFS an Euler-style tour of the trees to cover all cells within the $10n^2$ move limit.

Time complexity: $O(N^2)$

Problem H: Hoptimisation

Problem: Arrange up to 16 cell names along a line to minimise the total distance of a given sequence of hops.

Solution:

- → This is minimum linear arrangement; $k \leq 16$ makes an exponential subset DP feasible.
- → Rewrite total travel as the sum, over line boundaries, of transitions crossing each cut.
- → Use $\mathrm{dp}[S] = \min_{v \in S}(\mathrm{dp}[S \setminus \{v\}] + \mathrm{cutCost}[S])$ and retain choices to recover the order.

Time complexity: $O(N + 2^k k)$

Problem M: Mid Clocking

Problem: Place tasks in processor slots to minimise the finishing time when adjacent processors run more slowly.

Solution:

- → Binary-search a candidate makespan T.
- → A task with $t/0.6 > T$ needs zero neighbours; if $t/0.3 > T$, it can have at most one.
- → Check whether these task classes fit the optimal patterns of isolated slots, adjacent pairs, and dense 30% blocks.

Time complexity: $O(N \log N + \log(\text{range}/\varepsilon) \log N)$

Problem J: Jammer Jam

Problem: Assign swimmers to lanes to maximise their average speed when everyone in a lane ends up matching its slowest swimmer.

Solution:

- → Sort by speed; an optimal lane is a contiguous interval $[i, j]$ with total speed $(j - i + 1)a_i$.
- → Dynamic programming chooses interval boundaries for each lane, but the direct transitions are too slow.
- → Convexity/quadrangle structure enables divide-and-conquer or convex optimisation; Lagrange relaxation can reduce dependence on the lane count.

Time complexity: $O(N \log N)$ or $O(N \log(\text{range}))$

Problem D: Dire Strait

Problem: Find the shortest path between two vertices inside a simple polygon when a triangulation is given.

Solution:

- → The dual graph of the triangulation is a tree, so the endpoints have a unique triangle strip (the sleeve).
- → Treat the strip's interior edges as portals and tighten a visibility funnel through them.
- → When the funnel collapses, take the blocking corner and continue through the remaining portals.

Time complexity: $O(N)$